D 30573 **(Pages : 3)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2022

Mathematics

MTS 5D 01—APPLIED CALCULUS

(2020 Admission onwards)

Time : Two Hours Maximum : 60 Marks

Section A

Answer any number of questions.
Each question carries 2 *marks. Ceiling is* 20.

1. If $f(x) = 2x^2 - 3x + 1$, find $f(x-2)$.

2. Plot the points $(4,3), (-2,7), (5,-1)$ and $(-1,-8)$ in a rectangular co-ordinate plane.

3. Find $\lim_{x \to -1} \left(3x^3 - 4x + 8\right)$.

4. Differentiate the polynomial $y = x^6 - 3x^4 + 12x^2 - 8x + 5$.

5. Find the second derivative of $y = 2x^3 - 3x^2 + 12x - 8$.

6. Use logarithmic rules to expand $\log_2 \left(\frac{y^3}{x^5}\right)$.

7. Solve for $x : 3 = e^{20x}$.

8. Find $\int \left(3x^2 + 2x + 1\right) dx$.

9. Use the fundamental theorem of calculus to find the area of the region under the line $y = 2x + 1$ over the interval $1 \le x \le 3$.

Turn over

10. Find $\int_0^1 \left(4x^3 + 4\right) dx$.

11. Show that $ln \frac{1}{x} = - ln\, x$.

12. If $f\left(x\right) = 5^{x^2 + 2x,}$ find all values of x such that $f\left(x\right) = 125$.

Section B

Answer any number of questions.
Each question carries 5 *marks. Ceiling is* 30.

13. If $f\left(x\right) = x^2 + 3x + 1$ and $g\left(x\right) = x + 1$, find $f\left(g\left(x\right)\right)$ and $g\left(f\left(x\right)\right)$.

14. Graph the function $f\left(x\right) = x^2$.

15. Find :

(i) $\lim\limits_{x \to + \infty} \dfrac{\left(- x^3 + 2x + 1\right)}{x - 3}$; and

(ii) $\lim\limits_{x \to - 2} \dfrac{\left(x^2 - x - 6\right)}{x^2 + 3x + 2}$.

16. Verify the power rule for the function $\mathrm{F}\left(x\right) = \dfrac{1}{x^2} = x^{-2}$ by showing that its derivative is $\mathrm{F}'\left(x\right) = - 2x^{-3}$.

17. Find $\frac{dy}{dx}$ if $x^2 y + y^2 = x^3$.

18. Find all inflection points of $f\left(x\right) = 3x^5 - 5x^4 - 1$.

19. Evaluate $\int_0^1 8x \left(x^2 + 1\right)^3 dx$.

Section C

Answer any **one** *question.*
The question carries 10 *marks.*

20. Sketch the graph of the function $f(x) = \frac{x}{(x+1)^2}$.

21. Suppose \$1,000 is invested at an annual interest rate of 6%. Compute the balance after 10 years if the interest is compounded :

    a) Quarterly ;

    b) Monthly ;

    c) Daily ; and

    d) Continuously.

(1 × 10 = 10 marks)